© 2018 Human Kinetics, Inc.

Note: This article will be published in a forthcoming issue of the *International Journal of Sport Nutrition and Exercise Metabolism*. This article appears here in its accepted, peer-reviewed form; it has not been copyedited, proofed, or formatted by the publisher.

Section: Case Study

Article Title: Stress Relaxation and Targeted Nutrition to Treat Patellar Tendinopathy

Authors: Keith Baar

Affiliations: Department of Neurobiology, Physiology and Behavior, Department of Physiology and Membrane Biology, University of California Davis, Davis, CA.

Running Head: Loading and nutrition to treat tendinopathy

Journal: *International Journal of Sport Nutrition and Exercise Metabolism*

Acceptance Date: August 30, 2018

©2018 Human Kinetics, Inc.

DOI: https://doi.org/10.1123/ijsnem.2018-0231

"Stress Relaxation and Targeted Nutrition to Treat Patellar Tendinopathy" by Baar K
International Journal of Sport Nutrition and Exercise Metabolism
© 2018 Human Kinetics, Inc.

Stress Relaxation and Targeted Nutrition to Treat Patellar Tendinopathy

Keith Baar

Department of Neurobiology, Physiology and Behavior

Department of Physiology and Membrane Biology

University of California Davis

One Shields Ave

Davis, CA 95616

kbaar@ucdavis.edu

Downloaded by t_lambrinides@yahoo.com on 11/16/18

Running title: Loading and nutrition to treat tendinopathy

"Stress Relaxation and Targeted Nutrition to Treat Patellar Tendinopathy" by Baar K
International Journal of Sport Nutrition and Exercise Metabolism
© 2018 Human Kinetics, Inc.

Abstract

Patellar tendinopathy is one of the most common afflictions in jumping sports. This case study outlines the rehabilitation of a professional basketball player diagnosed by MRI with a central core patellar tendinopathy within the proximal enthesis. The player undertook a nutrition and strength based rehabilitation program combining gelatin ingestion and heavy isometric loading of the patella designed to produce significant stress relaxation as part of their competition schedule and a whole body training plan. On follow up one and a half years into the program an independent orthopedic surgeon declared the tendon normal on MRI. Importantly, the improved MRI results were associated with a decrease in pain and improved performance. This case study provides evidence that a nutritional intervention combined with a rehabilitation program that uses stress relaxation can improve clinical outcomes in elite athletes.

Keywords: Exercise, Tendon, Jumper's Knee, Loading, Rehabilitation

Downloaded by t_lambrinides@yahoo.com on 11/16/18

© 2018 Human Kinetics, Inc.

Downloaded by t_lambrinides@yahoo.com on 11/16/18

Introduction

Patellar tendinopathy (PT) is an extremely common musculoskeletal issue in elite athletes competing in jumping sports where reports of prevalence reach between 32% (basketball) and 44.6% (volleyball), whereas in body weight supported sports such as cycling the incidence of PT is low (Lian et al., 2005). Interestingly, sub-elite athletes report significantly lower patellar tendinopathy (14.4% for volleyball), suggesting that the volume or intensity of loading contributes to the development of this chronic problem (Hagglund et al., 2011; Janssen et al., 2015). Patellar tendinopathy is especially common in professional basketball due to the high playing load and intensity. In a preliminary report from the Australian National Basketball League, 52.3% of players reported patellar tendon pain that limited performance (Hannington et al., 2017), suggesting that the prevalence of this injury is high at the elite level.

Typically, severe chronic patellar tendinopathy can be visualized by MRI as an increase in signal intensity at the inferior pole of the patella (Johnson et al., 1996), suggesting that is develops from an injury to the proximal enthesis. In non-elite athletes, patellar tendinopathy is treated with rest and cryotherapy with limited success (Schwartz et al., 2015). Non-steroidal anti-inflammatory drugs (NSAIDs) are equivocal as to their effect on tendinopathy (Weiler, 1992). Indeed, recent work suggests that NSAIDs may not even be delivered to human tendons sufficiently to alter gene expression (Heinemeier et al., 2017).

One of the major issues surrounding patellar tendinopathy is that stress shielding can occur with loading (Almekinders et al., 2002), specifically at the insertion/origin regions (Orchard et al., 2004). Stress shielding is most commonly associated with the implantation of prosthetics within bone (Sumner, 2015). Essentially, the stronger metal components of the implant shield the neighboring bone from load (stress) resulting in more of the load being transmitted through the metal and subsequent loss of the "unloaded" native bone (Sumner, 2015). By analogy, healthy collagen can shield weaker or injured regions within the tendon resulting in unloading and a subsequent decrease in collagen content and orientation within the

© 2018 Human Kinetics, Inc.

injured region (Docking et al., 2016). The change in collagen content and orientation within the injured region is further associated with an increase in proteoglycans and water (Samiric et al., 2009), suggesting that the stress-shielded region of the tendon may be placed under a compressive load as the stronger collagen fibers squeeze (compress) the shielded region (Docking et al., 2013; Majima et al., 2000).

One way to overcome stress shielding is to take advantage of stress relaxation within the tendon (Atkinson et al., 1999). Stress relaxation is a fundamental property of viscoelastic tissues like tendons, where force within a tendon decreases over time when a static load is applied and the tendon is held at a constant length(Atkinson et al., 1999). Essentially, this means that when a tendon is loaded using isometric contractions the collagen fibers/fibrils/fascicles progressively relax over time, eventually reaching a steady state. As the collagen fibers/fibrils/fascicles relax, stress shielding within the tendon should decrease and a directional mechanical signal should be imparted to the injured region. Consistent with this hypothesis, isometric loading protocols decrease pain in high level athletes with tendinopathies (Rio et al., 2015; van Ark et al., 2016); however, whether a consistent program of isometric loading can improve the quality of the injured tendon has yet to be demonstrated.

On top of the positive effects of loading on tendon health, we have recently shown that nutrition can positively affect collagen synthesis in musculoskeletal tissues (Shaw et al., 2017). Briefly, we demonstrated that supplementing with 15 grams of gelatin, one hour before loading, resulted in an increase in whole body collagen synthesis and increased mechanics and collagen content of human engineered ligaments (Shaw et al., 2017). These data suggest that supplementing with gelatin can increase collagen synthesis. Improving collagen synthesis in stress shielded tendon might be important since immobilization for even short periods (10 days) has been shown to reduce tendon collagen synthesis by up to 50% and the decline in collagen synthesis continues out to 23 days (de Boer et al., 2007). Even though the decrease in collagen synthesis was not associated with a decrease in patellar tendon cross-

Downloaded by t_lambrinides@yahoo.com on 11/16/18

© 2018 Human Kinetics, Inc.

sectional area, the tendon did show decreased functional performance (de Boer et al., 2007), suggesting that decreased collagen synthesis is associated with impaired tendon function.

This paper, presents a case study of a professional basketball player with chronic patellar tendinopathy. The player was placed on a program that combined gelatin ingestion and stress relaxation loading as part of a rehabilitation protocol. Of note, even with an extensive playing and training schedule, the protocol significantly decreased MRI reactivity, increased the thickness of the mid-substance of the patellar tendon, and diminished pain. The loading and nutrition protocol presented here therefore could be one component of a rehabilitation protocol to improve functional outcomes in elite athletes.

Player and imaging

As part of the NBA scouting combine, all players invited to the event are given an MRI of both knees. For the athlete in question, one of his knees showed significant MRI reactivity at the inferior pole of the patella and MRI reactivity throughout the length of the tendon that was visible both in the sagittal T1-weighted images (Figure 1A) and axial PDW SPectral Attenuated Inversion Recovery (SPAIR) images (Figure 2A). At the time, the player was 21 years old and had experienced knee pain for ~5 years. The initial MRI was completed in May and the player was drafted that year and began working with the team 3 months later. Follow up MRIs were performed on a Philips Ingenia 3.0T MR system in August (Figures 1B and 2B) of the following year (15 months after the initial scan) and six months later (February of his second year in the league; Figures 1C and 2C) to gauge his progression. The initial MRI was completed on a separate machine, but great care was used to validate that equivalent images in the original and follow up scans were used for comparison. In the sections shown, the same cut was used and the patellar and tibial areas were similar.

Downloaded by t_lambrinides@yahoo.com on 11/16/18

© 2018 Human Kinetics, Inc.

Overview of interventions

After being informed of the purpose of the study and of all associated risks, the player gave written consent. The study was performed in accordance with the principles outlined in the Declaration of Helsinki and the University of California Davis Institutional Review Board.

In-Season Loading Volume

As the player was part of the active team, he participated in more than 50 games per season across two season averaging more than 20 minutes per contest. On top of game loading, he also participated in an average of 3 organized practices per week as well as a structured strength training program overseen by an elite strength and conditioning coach while being monitored by a rehabilitation specialist, physiotherapist, and sport medicine doctor.

Stress Relaxation Loading

The patellar tendon targeted training program consisted of mid-range isometric holds for 2 open chain (leg extension, and leg press) and one closed chain (Spanish squats) exercise. Both the weight lifted and the duration of the isometric hold progressively increased over the 18 months in the program. The isometric holds started at 10 seconds and progressively increased by 5 seconds and reached a peak of 30 seconds 6 months into the program. The stress relaxation program was performed twice a week through pre-season, in-season, and off-season workouts. The athlete was asked to perform 1-3 sets of 2-4 repetitions of single leg, mid-range isometric holds for leg extension, leg press, or Spanish squats. The holds were performed at >80% of 1 repetition max (1RM) and 1RM was evaluated every other month. The 30 second isometric holds were used because the force in patellar tendon sections drops by 60% by 30 seconds with only a further 10% decrease by 180 seconds (Atkinson et al., 1999). The length of the session (~10 minutes from start to finish) was used because we had previously found that short periods of loading

Downloaded by t_lambrinides@yahoo.com on 11/16/18

© 2018 Human Kinetics, Inc.

(5-10 minutes) were optimal for stimulating the cellular response to load and increasing collagen synthesis (Paxton et al., 2012).

Nutrition intervention

Normal dietary intake for the player was not tracked, nor did the player report changes in dietary intake over the length of the study. One hour before the patellar tendon targeted training program, the athlete consumed 15 grams of gelatin with ~225 mg vitamin C. The supplement consisted of commercial baking gelatin (Great Lakes unflavored beef gelatin) in 16oz of orange juice (containing ~225mg vitamin C). The amount of gelatin was based on our previous work showing that 15 grams of gelatin increased load-dependent collagen synthesis (Shaw et al., 2017). The timing of nutrients prior to exercise was designed to enhance nutrient delivery to the injured tendon and take advantage of cellular processes activated by the exercise stimulus to maximize collagen synthesis.

Compliance with the loading and nutrition program was greater than 95% in season and because of the early success the athlete maintained approximately this level of compliance in the offseason.

Body Composition Assessment

The player underwent body composition assessment via surface anthropometry using the sum of seven skinfolds at regular intervals throughout the program.

Strength Assessment

Isometric strength was determined using a Cybex 6000 isokinetic dynamometer calibrated before and after each test according to the manual.

Outcome of the intervention

Changes in body composition over the season (Table 1)

Over the course of the season, the player saw minimal changes in body mass (-2.5kg). The decrease in body weight corresponded with a decrease in fat mass (from 8.4% to 5.9%), indicating that a

Downloaded by t_lambrinides@yahoo.com on 11/16/18

© 2018 Human Kinetics, Inc.

large percentage of the body mass loss was the result of a decrease in fat mass. In agreement with this assessment, lean mass changed slightly over the testing period (75.6kg at the time of the initial MRI and 74.7kg at the time of the third MRI).

Functional progression through the program

In spite of a slight decrease in lean mass, maximal single leg isometric hamstring strength increased 196%, isometric leg extension strength increased by 156% and leg press strength increased 187.5% over the period of the program. The stress relaxation sets performed in each session for single legged isometric leg extensions progressed from 3 repetitions of 10 seconds with 12.5kg to 2 repetitions of 30 seconds with 32kg, whereas the single legged isometric leg press progressed from three 10 second holds with 80kg to two 30 second holds with 170kg.

Imaging and pain

Follow up MRI images were performed after a year in the program (15 months after the first MRI) and a further 6 months later, after the player had been on the program for ~18 months. The MRI images following 12 and 18 months into the program showed a progressive decrease in MRI reactivity at the inferior pole of the patella in both the sagittal T1 weighted images (Figure 1) and the axial SPAIR images (Figure 2). At the time of the third scan, a blinded orthopedic surgeon declared the patellar tendon normal. The reduction of reactivity at the inferior pole of the patella was accompanied by a decrease in the diameter of the patellar tendon at the proximal enthesis from 10.8mm at scan 1 to 8.7 and 8.16mm in scans 2 and 3, respectively. The decrease in the thickness of the proximal end of the patellar tendon was accompanied by an increase in the thickness at the mid-point of the tendon. The diameter of the midpoint of the patellar tendon progressively increased from 5.74 (scan 1), to 6.16 (scan 2) and then 6.29 (scan 3) mm and this was accompanied by a decrease in T1 and T2 reactivity (not shown). Therefore, the proximal end of the tendon decreased ~25%, whereas the middle of the tendon hypertrophied ~10%. Beyond the

Downloaded by t_lambrinides@yahoo.com on 11/16/18

© 2018 Human Kinetics, Inc.

changes in imaging, self-reported pain and tenderness within the tendon decreased. At the point of the final scan, the athlete was pain free, even though he was playing more than 25 minutes a game.

Reflections

The goal of this case study was to determine whether a targeted loading and nutritional program could enhance outcomes of a patellar tendinopathy rehabilitation program. Specifically, we used a loading program that maximized stress relaxation (long isometric holds) in the hopes of providing a uniaxial tensile load to the injured region of the patellar tendon. Together with the stress relaxation loading, we primed the athlete with 15 grams of gelatin containing 225mg of vitamin C so that once the cells were activated by the load they had would have the substrates necessary to increase collagen synthesis. The major results of the program were a decrease in pain and tenderness that corresponded with a significant decrease in MRI reactivity at the inferior pole of the patella and an increase in mid-substance patellar tendon thickness.

Stress relaxation is a fundamental property of viscoelastic materials like tendons and ligaments (Kjaer, 2004). Basically, as the tissue is loaded water is squeezed out and structures, such as fascicles and sub-fascicular components of the tendon slide relative to each other resulting in a progressive decrease in the tension within the tendon at a given length (Atkinson et al., 1999; Magnusson et al., 1997). In the human patellar tendon, two important observations about stress relaxation were first made by Atkinson and her colleagues (Atkinson et al., 1999). First, stress relaxation is bigger when larger parts of a tendon are loaded. This is important since it suggests that intrinsic differences in the strength of different fascicle or sub-fascicle regions are exposed during stress relaxation. In the current case, this is potentially important since it suggests that stronger regions of the tendon may relax early in the process and weaker/damaged regions of the tendon may be forced to take on more of the load as stress relaxation progresses. Since directional loading is fundamentally required for high degrees of collagen synthesis and matrix alignment (Kapacee et al., 2008), this suggests that stress relaxation can be used to provide a

Downloaded by t_lambrinides@yahoo.com on 11/16/18

© 2018 Human Kinetics, Inc.

tensile stimulus to damaged regions of the collagen matrix to promote directional collagen synthesis. The second important observation by Atkinson and colleagues (Atkinson et al., 1999) was that stress-relaxation reaches ~60% within 30 seconds and continues only a further 10% more up to 180 seconds. This suggests that heavy isometric holds lasting 30 seconds will be sufficient to provide a similar degree of stress relaxation as holds lasting 6-times longer. Therefore, the rationale for the loading paradigm was to use stress relaxation to provide a tensile load to the cells within the damaged central core of the proximal patellar tendon.

As important as tensile loading is to collagen synthesis, recent work from our group (Shaw et al., 2017) and others (Clark et al., 2008; McAlindon et al., 2011; Oesser et al., 1999) suggests that collagen synthesis can be enhanced by supplementing the diet with gelatin or hydrolyzed collagen. Specifically, we have previously shown that providing 15 grams of gelatin together with ~50mg of vitamin C an hour before loading resulted in an increase in collagen synthesis relative to a placebo or 5-gram dose of gelatin (Shaw et al., 2017). Therefore, for this case study we had the athlete consume 15 grams of gelatin within an orange juice slurry (containing ~225mg of vitamin C) one hour before the stress relaxation training. Even though we do not have a control group for this intervention, we have growing evidence that specific amino acids within the gelatin provide a stimulus to the cells of a tendon or ligament to increase collagen synthesis. Coupled with the directional tensile load, the added amino acids may be necessary to support the increase in collagen synthesis. However, much more work is needed before we can definitively understand the role of gelatin in the recovery described in this case study.

As with any intervention with elite athletes, the intervention can only be effective if there is buy in from the player. In this case, the injury was sufficiently painful that the player was willing to drink the slurry and perform the loading paradigm twice a week. When the treatment showed success, the player quickly became a major advocate of the program within the team. Currently, any player with symptoms of patellar or other forms of tendinopathy quickly opts into the program and now most members of the

Downloaded by t_lambrinides@yahoo.com on 11/16/18

© 2018 Human Kinetics, Inc.

team preform a form of prehab using the gelatin and stress relaxation loading along with their regular training. Even though we have not quantified musculoskeletal injuries before and after the intervention, reports suggest that visits to the training room and non-contact musculoskeletal injuries within the team have dropped.

Tendinopathies are multifactorial in nature and central core tendinopathies are particularly difficult to prevent and treat in an elite population. We hope that the program reported here can provide a starting place for treating these debilitating injuries.

Acknowledgements

The author wishes to thank the athlete who undertook this rehabilitation protocol and followed instructions diligently. Additionally, this project would not have been possible without the expertise, experience, and willingness of the Director of Strength and Conditioning and his staff, the Head Athletic Trainer and their Staff, and the Director of Player Health and Performance. The author is indebted to them for their efforts and willingness to publish the results. Lastly, thanks are owed to Dr. Letitia Bradford for independent evaluation of the blinded MRI images. The author has no conflicts of interest in regards to this case study. The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation, and do not constitute endorsement by ACSM.

Downloaded by t_lambrinides@yahoo.com on 11/16/18

© 2018 Human Kinetics, Inc.

References

Almekinders, L. C., Vellema, J. H., & Weinhold, P. S. (2002). Strain patterns in the patellar tendon and the implications for patellar tendinopathy. *Knee Surg Sports Traumatol Arthrosc, 10*(1), 2-5. doi:10.1007/s001670100224

Atkinson, T. S., Ewers, B. J., & Haut, R. C. (1999). The tensile and stress relaxation responses of human patellar tendon varies with specimen cross-sectional area. *J Biomech, 32*(9), 907-914.

Clark, K. L., Sebastianelli, W., Flechsenhar, K. R., Aukermann, D. F., Meza, F., Millard, R. L., . . . Albert, A. (2008). 24-Week study on the use of collagen hydrolysate as a dietary supplement in athletes with activity-related joint pain. *Current medical research and opinion, 24*(5), 1485-1496. doi:10.1185/030079908X291967

de Boer, M. D., Maganaris, C. N., Seynnes, O. R., Rennie, M. J., & Narici, M. V. (2007). Time course of muscular, neural and tendinous adaptations to 23 day unilateral lower-limb suspension in young men. *The Journal of Physiology, 583*(Pt 3), 1079-1091. doi:10.1113/jphysiol.2007.135392

Docking, S., Samiric, T., Scase, E., Purdam, C., & Cook, J. (2013). Relationship between compressive loading and ECM changes in tendons. *Muscles Ligaments Tendons J, 3*(1), 7-11. doi:10.11138/mltj/2013.3.1.007

Docking, S. I., & Cook, J. (2016). Pathological tendons maintain sufficient aligned fibrillar structure on ultrasound tissue characterization (UTC). *Scand J Med Sci Sports, 26*(6), 675-683. doi:10.1111/sms.12491

Hagglund, M., Zwerver, J., & Ekstrand, J. (2011). Epidemiology of patellar tendinopathy in elite male soccer players. *Am J Sports Med, 39*(9), 1906-1911. doi:10.1177/0363546511408877

Hannington, M., Rio, E., Padua, D., Stanley, L., Berkoff, D., Edwards, S., . . . Docking, S. (2017). Prevalence and impact of patellar tendinopathy on elite basketball athletes: Quantifying injury beyond the time-loss definition. *Journal of Science and Medicine in Sport, 20*(S3), 17-18.

Heinemeier, K. M., Ohlenschlaeger, T. F., Mikkelsen, U. R., Sonder, F., Schjerling, P., Svensson, R. B., & Kjaer, M. (2017). Effects of anti-inflammatory (NSAID) treatment on human tendinopathic tissue. *Journal of applied physiology, 123*(5), 1397-1405. doi:10.1152/japplphysiol.00281.2017

Janssen, I., Steele, J. R., Munro, B. J., & Brown, N. A. (2015). Previously identified patellar tendinopathy risk factors differ between elite and sub-elite volleyball players. *Scand J Med Sci Sports, 25*(3), 308-314. doi:10.1111/sms.12206

Johnson, D. P., Wakeley, C. J., & Watt, I. (1996). Magnetic resonance imaging of patellar tendonitis. *J Bone Joint Surg Br, 78*(3), 452-457.

Kapacee, Z., Richardson, S. H., Lu, Y., Starborg, T., Holmes, D. F., Baar, K., & Kadler, K. E. (2008). Tension is required for fibripositor formation. *Matrix Biol, 27*(4), 371-375.

Kjaer, M. (2004). Role of extracellular matrix in adaptation of tendon and skeletal muscle to mechanical loading. *Physiol Rev, 84*(2), 649-698.

Downloaded by t_lambrinides@yahoo.com on 11/16/18

© 2018 Human Kinetics, Inc.

Lian, O. B., Engebretsen, L., & Bahr, R. (2005). Prevalence of jumper's knee among elite athletes from different sports: a cross-sectional study. *Am J Sports Med, 33*(4), 561-567. doi:10.1177/0363546504270454

Magnusson, S. P., Simonsen, E. B., Aagaard, P., Boesen, J., Johannsen, F., & Kjaer, M. (1997). Determinants of musculoskeletal flexibility: viscoelastic properties, cross-sectional area, EMG and stretch tolerance. *Scand J Med Sci Sports, 7*(4), 195-202.

Majima, T., Marchuk, L. L., Sciore, P., Shrive, N. G., Frank, C. B., & Hart, D. A. (2000). Compressive compared with tensile loading of medial collateral ligament scar in vitro uniquely influences mRNA levels for aggrecan, collagen type II, and collagenase. *J Orthop Res, 18*(4), 524-531. doi:10.1002/jor.1100180403

McAlindon, T. E., Nuite, M., Krishnan, N., Ruthazer, R., Price, L. L., Burstein, D., . . . Flechsenhar, K. (2011). Change in knee osteoarthritis cartilage detected by delayed gadolinium enhanced magnetic resonance imaging following treatment with collagen hydrolysate: a pilot randomized controlled trial. *Osteoarthritis Cartilage, 19*(4), 399-405. doi:10.1016/j.joca.2011.01.001

Oesser, S., Adam, M., Babel, W., & Seifert, J. (1999). Oral administration of (14)C labeled gelatin hydrolysate leads to an accumulation of radioactivity in cartilage of mice (C57/BL). *J Nutr, 129*(10), 1891-1895.

Orchard, J. W., Cook, J. L., & Halpin, N. (2004). Stress-shielding as a cause of insertional tendinopathy: the operative technique of limited adductor tenotomy supports this theory. *J Sci Med Sport, 7*(4), 424-428.

Paxton, J. Z., Hagerty, P., Andrick, J. J., & Baar, K. (2012). Optimizing an intermittent stretch paradigm using ERK1/2 phosphorylation results in increased collagen synthesis in engineered ligaments. *Tissue engineering. Part A, 18*(3-4), 277-284. doi:10.1089/ten.TEA.2011.0336

Rio, E., Kidgell, D., Purdam, C., Gaida, J., Moseley, G. L., Pearce, A. J., & Cook, J. (2015). Isometric exercise induces analgesia and reduces inhibition in patellar tendinopathy. *Br J Sports Med, 49*(19), 1277-1283. doi:10.1136/bjsports-2014-094386

Samiric, T., Parkinson, J., Ilic, M. Z., Cook, J., Feller, J. A., & Handley, C. J. (2009). Changes in the composition of the extracellular matrix in patellar tendinopathy. *Matrix Biol, 28*(4), 230-236. doi:10.1016/j.matbio.2009.04.001

Schwartz, A., Watson, J. N., & Hutchinson, M. R. (2015). Patellar Tendinopathy. *Sports Health, 7*(5), 415-420. doi:10.1177/1941738114568775

Shaw, G., Lee-Barthel, A., Ross, M. L., Wang, B., & Baar, K. (2017). Vitamin C-enriched gelatin supplementation before intermittent activity augments collagen synthesis. *Am J Clin Nutr, 105*(1), 136-143. doi:10.3945/ajcn.116.138594

Sumner, D. R. (2015). Long-term implant fixation and stress-shielding in total hip replacement. *J Biomech, 48*(5), 797-800. doi:10.1016/j.jbiomech.2014.12.021

Downloaded by t_lambrinides@yahoo.com on 11/16/18

© 2018 Human Kinetics, Inc.

van Ark, M., Cook, J. L., Docking, S. I., Zwerver, J., Gaida, J. E., van den Akker-Scheek, I., & Rio, E. (2016). Do isometric and isotonic exercise programs reduce pain in athletes with patellar tendinopathy in-season? A randomised clinical trial. *J Sci Med Sport, 19*(9), 702-706. doi:10.1016/j.jsams.2015.11.006

Weiler, J. M. (1992). Medical modifiers of sports injury. The use of nonsteroidal anti-inflammatory drugs (NSAIDs) in sports soft-tissue injury. *Clin Sports Med, 11*(3), 625-644.

Downloaded by t_lambrinides@yahoo.com on 11/16/18

© 2018 Human Kinetics, Inc.

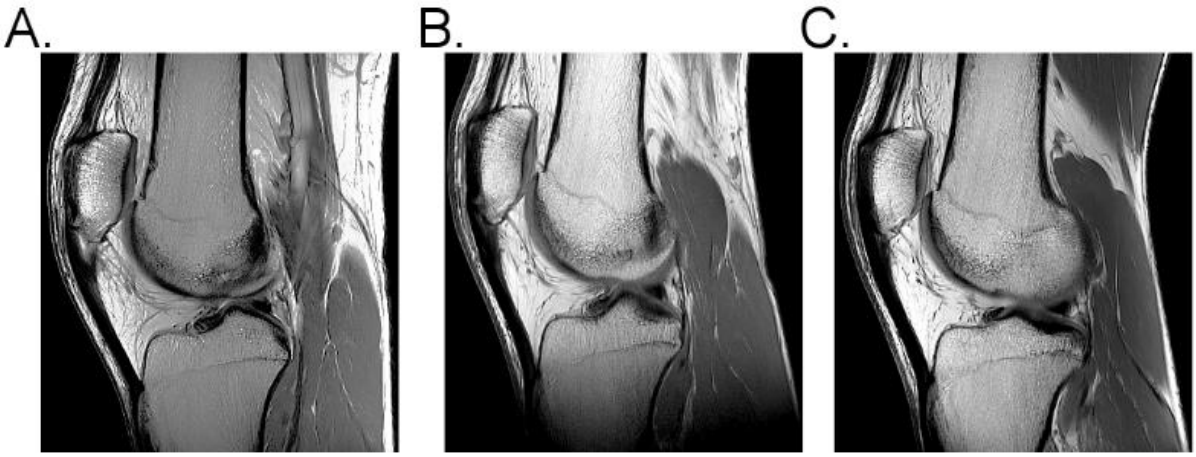

Figure 1. T1-weighted MRI images from A) before, B) 12 months, and C) 18 months into the intervention showing extent of the patellar tendinopathy within the central core of the inferior pole of the patella (note the white triangular region at the bottom of the patella in panel A) before the intervention and the progressive regression with the intervention.

Downloaded by t_lambrinides@yahoo.com on 11/16/18

© 2018 Human Kinetics, Inc.

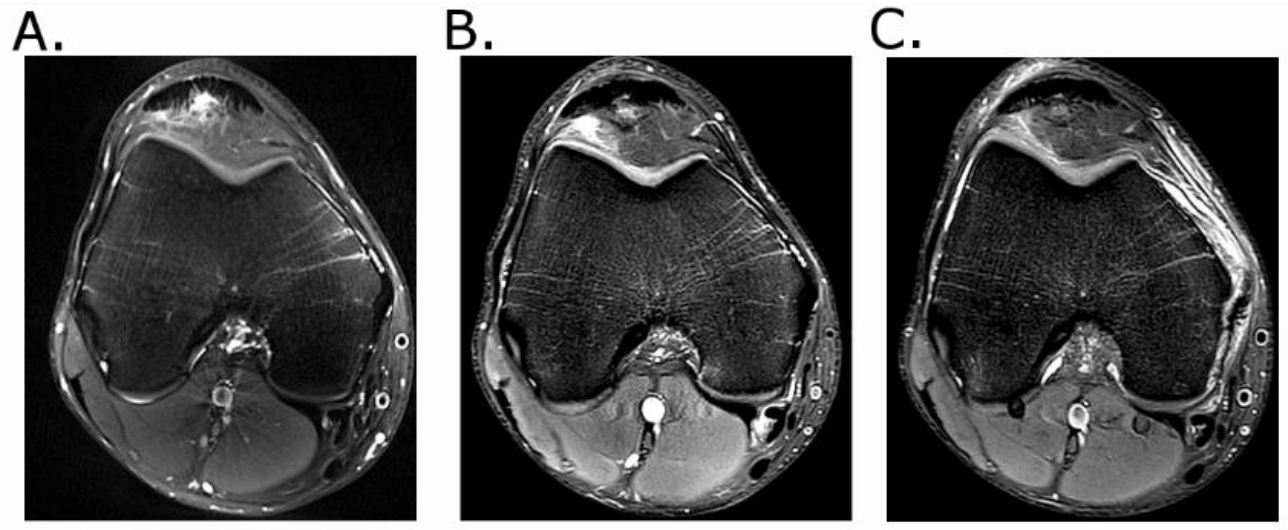

Figure 2. Axial PDW SPAIR MRI images from A) before, B) 12 months, and C) 18 months into the intervention showing the hole within the proximal patellar tendon before the intervention and the progressive loss of signal at the inferior pole of the patella following the intervention. Note also that the cross-sectional area of the patellar tendon is smaller after rehab.

Downloaded by t_lambrinides@yahoo.com on 11/16/18

© 2018 Human Kinetics, Inc.

Table 1: Anthropometric data across the intervention period

Month	Height (cm)	Weight (kg)	Skin Fold (mm)	% Body Fat
0	200	91.8	59.5	8.4
3	200	91.2	58.5	8.3
5	200	90.6	52	7.7
6	200	89.6	41.2	6.5
8	200	88.9	40.4	6.5
13	200	91.4	39	6.3
14	200	90.5	32.6	6.3
15	200	89.5	36.4	6.7
16	200	89.6	34.4	6.4
17	200	87.1	31	6.1
18	200	89.3	29.8	5.9

Downloaded by t_lambrinides@yahoo.com on 11/16/18